Broadband High Power Amplifier Module (SSPA)

REV.2.0_2024.11.29

6-18GHz/51dBm/Module

Model Number: KB60180M51A

The model KB60180M51A is a multi-octave high power amplifier operating between 6 GHz and 18 GHz and offering a wide dynamic Range with 51 dBm typical saturated power. The employment of gallium nitride (GaN) and chip-and-wire technology in manufacturing ensures this module state-of-the-art power performance with excellent power-to-volume ratio. It is ideal for jamming, EMC, test and measurement applications.

FEATURES:

- Solid-state design
- Instantaneous ultra-broadband
- 50Ohms input and Output matched
- Built-in protection circuits

ELECTRICAL SPECIFICATIONS @ +28.0VDC, 25°C, 50Ω

Parameter	Symbol	Minimum	Typical	Maximum	Units
Operating Frequency	BW	6		18	GHz
RF Output Power @$P_{in=0dBm}$	P_{SAT}		51		dBm
Power Gain @P_{SAT}	Gp		51		dB
Power Gain Flatness	Δ Gp		±2		dB
Input Return Loss	S_{11}		-10		dB
Harmonics @100W	H		-15		dBc
Spurious Signals	Spur		-55		dBc
In/Output Impedance	Impedance		50		Ω
Operating Voltage	V_{DC}	26	28	30	Volt
DC Current	I_{DD}		42	48	Amp

MECHANICAL SPECIFICATIONS

Parameter	Value	Units	Limits
Dimensions	260x216x110[10.24x8.5x4.33]	mm [inch]	
Weight	8 [3.04]	kg [lbs]	Maximum
RF Connectors Input	SMA, Female		
RF Connectors Output	N, Female		
DC Interface Connector	J29A-37ZKP		
Cooling	With heatsink, forced air cooling		

ENVIRONMENTAL CHARACTERISTICS (Design to Meet)

Parameter	Minimum	Typical	Maximum	Units	Notes
Operating Temperature	-40		60	°C	
Non-operating Temperature	-45		85	°C	Storage
Relative Humidity (non-condensing)	5		95	%	

ABSOLUTE MAXIMUM RATING

Input RF drive level without damage	+5dBm	Maximum
Load VSWR @ POUT =80W	2.5:1 @ all load phase & amplitude continuous More than 3:1 may cause PA damage	
Over Temperature	80°C @ heatsink	

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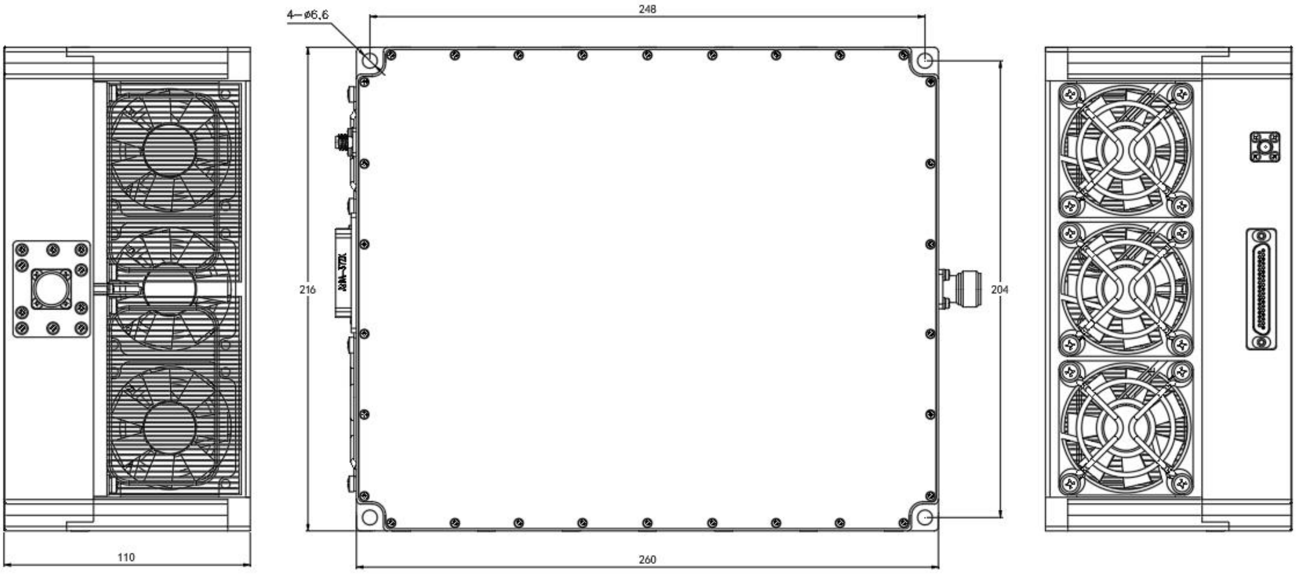
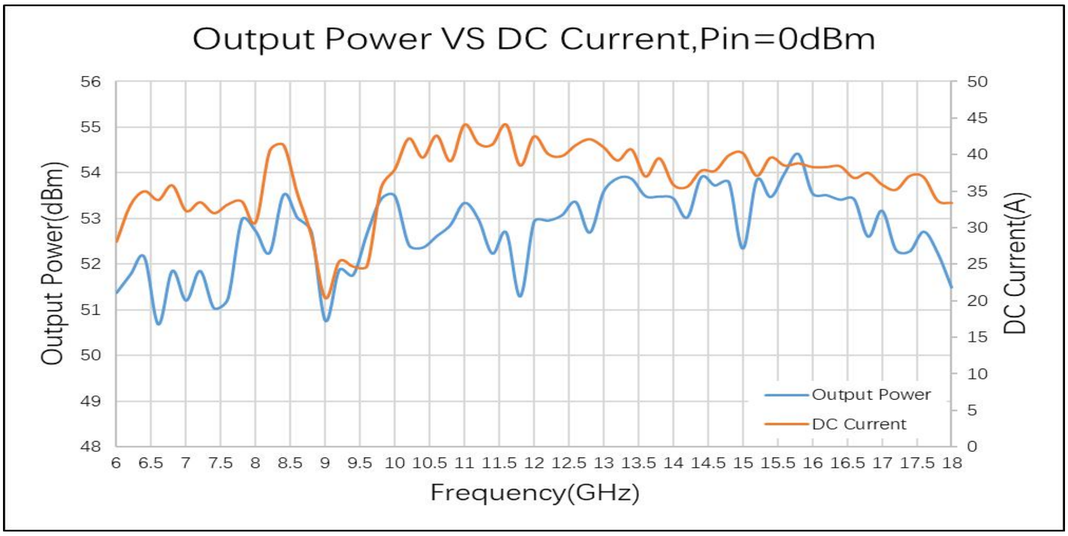
DC INTERFACE CONNECTOR

Pin #	Description	Specifications
1-15	VDD	28 V
16-31	GND	Ground
32	ENABLE	Amplifier Enable: TTL Logic High (3.3V)
33	CURRENT SENSE	Analog voltage relative to IDD
34	CURRENT ALARM	TTL Logic High (3.3V), when current was exceed limit value
35	TEMP ALARM	TTL Logic High (3.3V) , when temperature was exceed limit value
36-37	GND	Ground

OUTLINE DRAWING [All dimensions in mm)]



TYPICAL PERFORMANCE PLOTS (For reference only)



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